

KAUSTUBH DWIVEDI

Aerospace Structures & Design Engineer

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PROFILE

I am an aerospace structures engineer who has just completed an MSc at Cranfield University. I do not treat CAD, simulation and testing as separate tasks: a design is not finished for me until the model, the manufacturing constraints and the test evidence agree, and I say plainly when they do not. I am the person on a team who wants to understand why something works before trusting it, who keeps the plan and the documentation straight so others can pick up my work, and who takes feedback from more experienced engineers as the fastest way to get better. I have coordinated the technical work of a student team, worked to an industrial review cadence and published with co-authors, and what I want next is a UK structures, design or analysis role where I can learn from experienced engineers and be useful from the first week.

EXPERIENCE

Airbus A320 Slat-Track Can — Additive Manufacturing Redesign and Structural Substantiation

2025 – 2026

Supervised and reviewed by GKN Aerospace · MSc research project, Cranfield University

- **Released a monolithic additively manufactured AlSi10Mg design** at 1.478 kg in place of a multi-part welded 6061-T6 assembly, removing 20.6 % of the mass with all five load cases elastic and a minimum yield factor of safety of 5.56.
- **Quantified what the redesign actually bought:** at a fixed 1.450 kg, redistributing wall material alone cut displacement 3.4–5.1 % and von Mises stress 3.7–10.1 %, and a uniform 1.50 mm wall of comparable mass yielded locally where the variable wall stayed elastic.
- **Set the load basis with the industrial supervisor** where the certified definition was proprietary — a five-case directional pressure matrix on CS 25.963 fuel-head loading, bounded against CS 25.561 emergency-landing inertia.
- **Down-selected the manufacturing route** across four AM candidates against the existing welded design, rejecting both polymer routes on quantified stiffness grounds rather than on preference, and carrying the variable-wall method forward to the metal design.
- **Released the geometry as a watertight parametric CATIA V5 solid**, exported as a STEP file and re-verified on an independent 239,329-element C3D10 mesh sharing no nodes with the development model.
- **Screened the existing welded can for fatigue** to EN 1999-1-3 detail category 50–4.3 in nCode: 1.145×10^6 repeats and a 3.82× margin on the 300,000-repeat target, cross-checked to within 3 % by an independent shell tangent-stress route.
- **Assessed manufacturing risk** with ANSYS Additive thermal simulation, identifying the root and collar band as the sustained hotspot and establishing from a control run on the existing welded geometry that the pattern follows build direction, not the redesign.
- **Closed the work out against a requirements verification matrix**, separating what the numerical evidence closes from what still needs supplier allowables, a buckling check and leak/proof testing before prototype release.

Composite Wing Leading Edge — Bird-Strike Concept Assessment

2025 – 2026

GKN Aerospace / Cranfield University · industry-linked group design project · technical coordination

- **Ranked five leading-edge concepts** — CFRP, GFRP, Kevlar/epoxy, honeycomb sandwich and metallic — under common loading, using Abaqus/Explicit SPH-FEA at 90 and 180 m/s soft-body impact on deformation, peak force, stress distribution and energy absorption.
- **Validated the models against specimens the team built and tested**, working closely with teammates to reconcile simulation and test results until the whole group could trust the conclusions.
- **Converted the analysis into design recommendations**, weighing laminate behaviour and manufacturability alongside impact performance.
- **Ran the technical coordination:** task allocation, deadline tracking, integration of teammates' analysis and test inputs, and presentation of recommendations at project reviews. The study is published open access in the Journal of Manufacturing and Materials Processing (2026).

Engineering Design Intern — La Fondation Dassault Systèmes / Dassault Systèmes

Dec 2022 – May 2024

Pune, India · 18-month industry placement

- **Took a modular multi-ratio gearbox from concept to working prototype**, combining CATIA V5 / 3DEXPERIENCE modelling, hand calculation and Abaqus assessment.
- **Iterated the design against load transfer, packaging and manufacturability constraints**, keeping the technical documentation current with each change.
- **Worked to a structured review cadence** inside a Dassault Systèmes engineering environment, responding to senior-engineer feedback and managing design changes through to sign-off.

TECHNICAL SKILLS

Finite element	Abaqus/Standard and /Explicit — shell and solid modelling, geometric and material nonlinearity, contact and tie constraints, SPH impact, mesh sensitivity studies, verification against an independent mesh, simulation–test correlation.
Optimisation & AM	Tosca Structure (bead and shape), ANSYS Additive thermal simulation; DfAM — wall thickness, build orientation, support strategy, process risk.
CAD & release	CATIA V5 and 3DEXPERIENCE — part, assembly and multi-section surface modelling, parametric solid reconstruction, STEP export and import, geometry validation and design documentation.
Composites	Laminate and sandwich design, stacking sequence definition, specimen manufacture, mechanical and impact testing, damage and delamination modelling.
Standards & method	CS 25.963 and CS 25.561 load definitions; EN 1999-1-3 fatigue curves; nCode; requirements verification matrices and evidence chains.
Programming	Python for post-processing, mesh and field-data handling and figure generation; MATLAB; Excel.

EDUCATION

MSc Advanced Lightweight and Composite Structures — Cranfield University Accredited by RAeS and IMechE for further learning toward CEng registration. Composite structures, finite element methods, structural stability, impact mechanics. <i>Thesis: Slat Track Cans of the Future — Exploring the Design Space of a High Criticality Fuel-Tank Component.</i>	Sept 2025 – Sept 2026
B.Tech Mechanical Engineering — Symbiosis Institute of Technology, Pune CGPA 8.36 / 10.	2020 – 2024

PUBLICATIONS AND RESEARCH

Lalehparvar, M., Nuttall, A., Bavaria, D., Massó Etxeberria, F., Dwivedi, K., Ghasemnejad, H., Coladas Mato, P. and van de Waerdt, W. (2026). Comparative SPH–Finite Element Assessment of Aerospace Material Systems Under Bird-Strike Loading. *Journal of Manufacturing and Materials Processing*, 10(9), 343. Open access. doi:10.3390/jmmp10090343

Dwivedi, K., et al. (2024). Optimizing 3D Printed Diamond Lattice Structure and Investigating the Influence of Process Parameters on their Mechanical Integrity Using Nature-Inspired Machine Learning Algorithms. *Materials Today Communications*, 38, 108233. doi:10.1016/j.mtcomm.2024.108233

Co-author of three peer-reviewed chapters (doi:10.1201/9781003437369) in *Sustainable Materials: The Role of Artificial Intelligence and Machine Learning*, CRC Press, 2024, on polymer welding, process optimisation and machine-learning classification of 3D-printed parts.

- Designed, manufactured and compression-tested 3D-printed PLA+ diamond lattice specimens, linking process parameters and cell geometry to mechanical performance.

MEMBERSHIP AND STATUS

- Royal Aeronautical Society — Student Affiliate. Issued Nov 2025, membership ID 3072018.
- Right to work in the UK and eligible for the Graduate visa, so no sponsorship is required. Open to UK relocation, hybrid working and travel to client or manufacturing sites.